

Program: Diploma in Cloud Computing and Big data	
Course Code: 3277	Course Title: Digital Computer Architecture Lab
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Interpret the principles and methodology of digital logic design.
- Identify the various digital ICs and understand their operation
- Build, analyze, and troubleshoot simple combinational and sequential circuits.
- Develop basic assembly language programs in 8085.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Knowledge of basic electrical and electronics engineering		Basics of Electrical and Electronics	2

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate realization of digital circuit using gates	10	Applying
CO2	Implement canonical and standard boolean functions, combinational logic circuits	12	Applying
CO3	Demonstrate flip flops and registers	10	Applying
CO4	Develop assembly language programs for 8085 microprocessor	10	Applying
	Lab Test	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2	2				
CO3	3	2	2				
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Demonstrate the basic digital circuit component-gates ,the concept of universal gates		
M1.01	Demonstrate the Logic gates - AND,OR,NOT,NAND,NOR,EXOR ,EXNOR	5	Applying
M1.02	Implement basic gates using universal gates.	5	Applying
CO2	Implement canonical and standard boolean functions , combinational logic circuits		
M2.01	Use K Map for simplification into SOP and POS expression	3	Applying
M2.02	Demonstrate Half Adder and Full Adder.	3	Applying
M2.03	Implement Encoder and Decoder	3	Applying
M2.04	Construct Multiplexer and Demultiplexer using logic gates.	3	Applying
	Lab Test – I	3	
CO3	Demonstrate flip flops and registers		

M3.01	Demonstrate Flip Flops: SR, D, T, JK Flip Flops	4	Applying
M3.02	Implement 4 bit shift register SISO,SIPO using D flip flop	4	Applying
M3.03	Use D Flip Flops to implement PISO,PIPO shift registers	2	Applying
C04	Develop assembly language programs in 8085 microprocessor		
M4.01	Familiarization of 8085 Microprocessor simulator	2	Understanding
M4.02	Implement DataTransfer instructions	2	Applying
M4.03	Demonstrate arithmetic instructions.	2	Applying
M4.04	Illustrate Execution of Logic instructions	2	Applying
M4.05	Implement Branch instructions	2	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	M. Morris Mano & Michael D. Ciltti, Digital Design, Pearson Education, 5th Edition
T2	A. Anand Kumar, Fundamentals of digital circuits, PHI Learning Pvt. Ltd., 2003
T3	Ramesh Gaonkar ,Microprocessor architecture ,Programming,and applications with 8085,5th Ed.
R3	Malvino & Leach, Digital Principles and Applications, McGraw-Hill

Online Resources

Sl.No	Website Link
1	https://www.iitg.ac.in/cseweb/vlab/Digital-System-Lab/experiments.php
2	https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/
3	https://www.vlab.co.in/broad-area-electronics-and-communications
4	http://www.asic-world.com/digital/tutorial.html
5	https://en.wikipedia.org/wiki/Digital_electronics